

Atlas Reaper

Software Engineer / AI Infrastructure & Audio Systems

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Profile

Final-year BA (Hons) Game Design and Production student at Abertay University and Junior Software Engineer at Simple Online Healthcare. Work spans production web engineering, automated testing, CI/release tooling, local AI infrastructure, and real-time audio systems. Selected through the 2026 Saltire Scholars Programme and retained part-time through final year, with a full-time pathway after graduation.

Founder of Atlas Systems, an independent engineering estate spanning local AI, automation, infrastructure, observability, retrieval systems, and interactive audio. Built SONIN, a dual-engine generative audio-visual system awarded 100% with faculty recognition for advanced DSP awareness and architectural maturity.

Experience

- Jun 2026–Present **Junior Software Engineer, Simple Online Healthcare, Remote**
- Contribute to production-facing Next.js/TypeScript healthcare journeys, including member-facing flows, supporting frontend implementation, bug fixing, API integration, and cross-platform debugging.
 - Expanded automated quality coverage using Vitest, React Testing Library, Playwright, and synthetic journey testing to improve confidence in member-facing flows and regressions.
 - Built reusable developer tooling and documented scenario-based development and testing patterns to make realistic local testing faster and more consistent.
 - Diagnosed and resolved CI/release failures across Changesets, GitHub Actions, and GitHub Packages, tracing issues through release metadata to verified publication.
 - Authored internal AI skills, guidance, and knowledge-base material to make AI-assisted development more consistent, safer, and easier to adopt across projects.
- Sept 2023–Present **Founder & Systems Engineer, Atlas Systems, Independent**
- Design and operate a governed technical estate spanning local AI, automation, infrastructure, observability, real-time audio, and 35 governed public repositories at atlas-systems.uk.
 - Built reusable GitHub Actions and Cloudflare delivery paths separating pull-request validation, isolated browser-evidence previews, production deployment, and post-deployment verification.
 - Built a local AI knowledge layer using FastAPI, Ollama, ChromaDB, type-aware chunking, hybrid vector/BM25 retrieval, cited answer synthesis, and explicit public/private source boundaries.
 - Developed interactive engineering systems including System Symphony, Spectral Forge, telemetry interfaces, architecture explorers, and observability surfaces backed by real estate data and deterministic evidence.
- Jun 2026–Present **Saltire Scholar, Entrepreneurial Scotland, Global**
- Selected through Scotland's national Saltire Scholars Programme. The placement at Simple Online Healthcare led to continued part-time employment through final year and a full-time pathway after graduation.
- Serve as a lead liaison between scholars and the programme team, helping coordinate communication, feedback, and engagement across the cohort.
 - Act as a Saltire brand ambassador, representing the programme and its values across professional, university, and external-facing settings.

Selected Engineering Projects

- 2026 **System Symphony, Atlas Systems**
- Built a read-only generative instrument that translates bounded infrastructure telemetry into deterministic, replayable, and auditable real-time music, with explicit treatment of unknown and unmeasured states.
- 2026 **SlamPunk: 15-Stem Dynamic Mix Engine, Abertay University**
- Designed and implemented a real-time dynamic audio system for competitive play, coordinating 15 stems with game-state-driven mix behaviour in Unreal Engine 5.

2025 **SONIN: Dual-Engine Generative Audio-Visual System**, *Abertay University*, **Grade: 100%**

The only cohort submission cited by faculty for “advanced DSP awareness” and “architectural maturity.”

- Designed a dual-engine Max/MSP architecture for real-time algorithmic audio and visual generation.
- Engineered fault-tolerant data flow and compensation logic across independently operating systems.

Education

2023–2027 **BA (Hons) Game Design and Production**, *Abertay University*, Dundee

Industry-accredited games programme with a technical focus on interactive systems, audio implementation, production, and systems design.

- **Final-year focus:** interactive systems, technical audio, AI-assisted workflows, software architecture, and production engineering.

2019–2021 **Aeronautical Engineering – Undergraduate Coursework**, *University of Glasgow*, Glasgow

Coursework included Thermodynamics, Fluid Mechanics, Engineering Mathematics, and Structural Analysis before transferring into games and interactive systems.

2013–2019 **Advanced Highers & Highers**, *Dollar Academy*

Advanced Highers: Mathematics (A), Physics (A), Engineering Science (A).

Technical Skills

Languages Python, TypeScript/JavaScript, PHP, C++, C#

Web React, Next.js, REST APIs, HTML/CSS

AI / Retrieval Ollama, Open WebUI, ChromaDB, RAG, hybrid vector/BM25 retrieval

Local model evaluation, OpenAI and Anthropic API integration

Testing Vitest, React Testing Library, Playwright, synthetic journey testing

Platform GitHub Actions, Changesets, GitHub Packages, Docker

Cloudflare Workers, Pages, KV, D1, AWS, DNS/TLS, Git, Perforce

Audio Unreal Engine 5, MetaSounds, Max/MSP, DSP, FL Studio, REAPER

Production Agile/Scrum/Kanban, Jira, technical documentation

Recognition

2026 **Saltire Scholar**, *Entrepreneurial Scotland*

Selected through Scotland's national Saltire Scholars Programme.

2019 **Academic Excellence**, *SQA Advanced Highers*

Grade A across Mathematics, Physics, and Engineering Science.

Selected Technical Writing

W-08 | SPECULAR-CORE: Architectural Recovery

Architectural recovery of a live technical estate, documenting boundaries, failure modes, governance, and evidence.

W-06 | Atlas Systems CI/CD

Design and assurance of a controlled delivery pipeline with explicit validation, deployment, and verification stages.

W-03 | Ramone: Local AI System

Local AI architecture covering retrieval, reproducible infrastructure, grounded responses, and operational boundaries.

Earlier Experience

May **F&B Supervisor**, *Crieff Hydro Hotel*, Crieff

2022–Apr Assumed full operational responsibility during an extended management absence and helped move guest satisfaction from the lowest-rated to the second-highest-rated department in the hotel during the same period.

Jun 2021–Apr **Team Manager**, *Hunter Sales and Co.*, Glasgow

2022 Promoted within the first quarter; built and led a B2B sales team, introducing structured conflict-resolution and planning processes that improved team output.